

Listeriolysin O secreted by *Listeria monocytogenes* induces NF-kappaB signalling by activating the IkappaB kinase complex.

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Listeriolysin O (LLO) is a pore-forming cytolysin secreted by the pathogen *Listeria monocytogenes* and is required for its intracellular survival. We recently demonstrated that in endothelial cells, LLO activates the NF-kappaB signalling pathway. In this work, we studied the LLO-induced molecular cascade of NF-kappaB activation with a cellular model extensively used to analyse the signalling pathway of NF-kappaB activation, i.e. the human embryonic kidney HEK-293 cell line and its derivatives (transfectants or mutants). When the stably transfected derivative HEK-293 cells expressing IL-1RI were exposed to LLO, a strong NF-kappaB activation was detected, contrasting with other cell lines (HEK-293 wild type, HEK-293.T and COS) expressing a very low level of IL-1RI. Although a delayed kinetics of LLO-dependent NF-kappaB activation suggests an autocrine or paracrine IL-1-dependent pathway, we found that LLO-dependent NF-kappaB activation did not require the IL-1 protein synthesis nor the interaction with the IL-1RI specific receptor. Herein, we demonstrated that LLO-dependent NF-kappaB activation requires the activation of the IkappaB kinase beta (IKKbeta) subunit of IKK complex to phosphorylate and degrade cytoplasmic IkappaBalpha, a natural inhibitor of NF-kappaB. The activation induced by LLO does not require the adapters MyD88 and IL-1R-associated kinase (IRAK). We suggested that LLO induces a distinct signalling pathway from that of IL-1 and its receptor.

Mechanisms of processing and presentation of the antigens of *Listeria monocytogenes*.

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Antigen processing is the series of events through which protein antigens become degraded and are sent to the cell surface for recognition by T cells. These events within the cell have been studied extensively using the model system of infection with *Listeria monocytogenes*. This bacteria resides primarily intracellularly; immunity to it is mediated by cellular responses. Upon phagocytosis by a macrophage, engulfed *Listeria* express a hemolytic molecule, listeriolysin O (LLO), and can disrupt the membrane of the endosome and escape into the cellular cytoplasm. This allows the organism to escape the hostile environment of the endosome and also gives it access to the processing machinery of the cytoplasm. Therefore, *Listeria monocytogenes* can be processed within an endosome and be presented by class II MHC, or can escape into the cytoplasm and be processed there and presented by class I MHC. This LLO molecule is not only an important virulence factor, it is also a dominant antigen in the cellular immune response to *Listeria*. The outcome of antigen processing and presentation can be influenced by the expression of LLO, by the state of activation of the macrophage, and by the cytokines involved in the immune response.

Identification of a PEST-like motif in listeriolysin O required for phagosomal escape and for virulence in *Listeria monocytogenes*.

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The hly-encoded listeriolysin O (LLO) is a major virulence factor secreted by the intracellular pathogen *Listeria monocytogenes*, which plays a crucial role in the escape of bacteria from the phagosomal

compartment. Here, we identify a putative PEST sequence close to the N-terminus of LLO and focus on the role of this motif in the biological activities of LLO. Two LLO variants were constructed: a deletion mutant protein, lacking the 19 residues comprising this sequence (residues 32-50), and a recombinant protein of wild-type size, in which all the P, E, S or T residues within this motif have been substituted. The two mutant proteins were fully haemolytic and were secreted in culture supernatants of *L. monocytogenes* in quantities comparable with that of the wild-type protein. Strikingly, both mutants failed to restore virulence to a hly-negative strain *in vivo*. *In vitro* assays showed that *L. monocytogenes* expressing the LLO deletion mutant was strongly impaired in its ability to escape from the phagosomal vacuole and, subsequently, to divide in the cytosol of infected cells. This work reveals for the first time that the N-terminal portion of LLO plays an important role in the development of the infectious process of *L. monocytogenes*.

Subversion of cellular functions by *Listeria monocytogenes*.

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Listeria monocytogenes is a Gram-positive bacterium that is able to invade and multiply within eukaryotic cells. Its intracellular life-cycle includes pathogen-induced phagocytosis, lysis of the phagocytic vacuole, movement in the cytoplasmic environment, and a cell-to-cell spread mechanism. Many *L. monocytogenes* virulence factors have been studied in detail, certain of which subvert specific eukaryotic cell functions in order to favour infection. During entry, the invasion protein InlA takes advantage of the adhesion molecule E-cadherin and the adherens junction machinery to adhere to target and invade polarized epithelial cells. Another invasion protein of the internalin family, InlB, subverts the signalling pathway of the hepatocyte growth factor receptor Met to induce endocytosis of the receptor and also to favour internalization of the bacteria in non-polarized epithelial cells. Once inside the cell, the haemolysin of *L. monocytogenes*--the listeriolysin O or LLO--is secreted to lyse the phagocytic vacuole, and when the bacteria is freed in the cytoplasm, the activity of the LLO is in part regulated by the infected cell itself, taking advantage of the pH sensitivity of the LLO that leads to its inactivation in the neutral eukaryotic cell cytoplasm. Finally, to induce bacterial movement in the cytoplasm, the *L. monocytogenes* surface protein ActA mimics the activity of the eukaryotic WASP family of proteins to recruit to the bacteria the actin nucleation machinery required for actin polymerization and for the formation of the actin structures (called 'actin comet tails') that propel the parasite in the cytosol and help it to invade neighbouring cells.

Promyelocytic Leukemia Protein (PML) Controls *Listeria monocytogenes* Infection.

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The promyelocytic leukemia protein (PML) is the main organizer of stress-responsive subnuclear structures called PML nuclear bodies. These structures recruit multiple interactors and modulate their abundance or their posttranslational modifications, notably by the SUMO ubiquitin-like modifiers. The involvement of PML in antiviral responses is well established. In contrast, the role of PML in bacterial infection remains poorly characterized. Here, we show that PML restricts infection by the pathogenic bacterium *Listeria monocytogenes* but not by *Salmonella enterica* serovar Typhimurium. During infection, PML undergoes oxidation-mediated multimerization, associates with the nuclear matrix, and becomes de-SUMOylated due to the pore-forming activity of the *Listeria* toxin listeriolysin O (LLO). These events trigger an antibacterial response that is not observed during *in vitro* infection by an LLO-defective *Listeria*

mutant, but which can be phenocopied by specific induction of PML de-SUMOylation. Using transcriptomic and proteomic microarrays, we also characterized a network of immunity genes and cytokines, which are regulated by PML in response to *Listeria* infection but independently from the listeriolysin O toxin. Our study thus highlights two mechanistically distinct complementary roles of PML in host responses against bacterial infection. The promyelocytic leukemia protein (PML) is a eukaryotic protein that can polymerize in discrete nuclear assemblies known as PML nuclear bodies (NBs) and plays essential roles in many different cellular processes. Key to its function, PML can be posttranslationally modified by SUMO, a ubiquitin-like modifier. Identification of the role of PML in antiviral defenses has been deeply documented. In contrast, the role of PML in antibacterial defenses remains elusive. Here, we identify two mechanistically distinct complementary roles of PML in antibacterial responses against pathogens such as *Listeria*: (i) we show that PML regulates the expression of immunity genes in response to bacterial infection, and (ii) we unveil the fact that modification of PML SUMOylation by bacterial pore-forming toxins is sensed as a danger signal, leading to a restriction of bacterial intracellular multiplication. Taken together, our data reinforce the concept that intranuclear bodies can dynamically regulate important processes, such as defense against invaders.

Exploring the role of the CTL epitope region of listeriolysin O in the pathogenesis of *Listeria monocytogenes*.

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Listeria monocytogenes is a facultative intracellular bacterial pathogen responsible for severe opportunistic infections in humans and animals. The secreted cholesterol-dependent cytolysin, listeriolysin O (LLO), mediates phagosomal escape and allows bacterial growth in the cytosol of infected cells. In order to identify new LLO determinants participating in bacterial pathogenesis, this study focused on a major target of LLO proteolytic cleavage in vitro, the CTL epitope region (residues 91-99). Mutations were generated by site-directed mutagenesis in the epitope or in the two clusters of positive charges flanking the epitope. Two LLO mutants (a single mutation K103A and a double mutation R89G, K90G) were normally and stably secreted by *L. monocytogenes*. In contrast, a mutant carrying four amino acid substitutions in the epitope itself (Y92K, D94A, E97K, Y98F) was highly susceptible to proteolytic degradation. While these three LLO mutant proteins showed a reduced haemolytic activity, they all promoted efficient phagosomal escape and intracellular multiplication in different cell types, and were non-cytotoxic. The deletion of the epitope (Delta91-99), as well as the substitution of two, three or four of the four lysine residues (K103 to K106) by alanine residues did not lead to the production of a detectable protein. These results confirm the lack of correlation between haemolytic activity and phagosomal membrane disruption. They reveal the importance of the 91-99 region in the production of a stable and functional LLO. LD(50) determinations in the mouse model suggest a possible link between LLO stability and virulence.

Listeriolysin O as a strong immunogenic molecule for the development of new anti-tumor vaccines.

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The pore-forming toxin listeriolysin O (LLO), which is produced by *Listeria monocytogenes*, mediates

bacterial phagosomal escape and facilitates bacterial multiplication during infection. This toxin has recently gained attention because of its confirmed role in the controlled and specific modulation of the immune response. Currently, cancer immunotherapies are focused on conquering the immune tolerance induced by poorly immunogenic tumor antigens and eliciting strong, lasting immunological memory. An effective way to achieve these goals is the co-administration of potent immunomodulatory adjuvant components with vaccine vectors. LLO, a toxin that belongs to the family of cholesterol-dependent cytolysins (CDCs), exhibits potent cell type-non-specific toxicity and is a source of dominant CD4(+) and CD8(+) T cell epitopes. According to recent research, in addition to its effective cytotoxicity as a cancer immunotherapeutic drug, the non-specific adjuvant property of LLO makes it promising for the development of efficacious anti-tumor vaccines.

Avoiding death by autophagy: interactions of *Listeria monocytogenes* with the macrophage autophagy system.

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Autophagy restricts the growth of a variety of intracellular pathogens. However, cytosol-adapted pathogens have evolved ways to evade restriction by this innate immune mechanism. *Listeria monocytogenes* is a Gram-positive bacterial pathogen that utilizes a cholesterol-dependent pore-forming toxin, listeriolysin O (LLO), to escape from the phagosome. Autophagy targets *L. monocytogenes* in LLO-damaged phagosomes and also in the cytosol under some experimental conditions. However, this bacterium has evolved multiple mechanisms to evade restriction by autophagy, including actin-based motility in the cytosol and an as yet undefined mechanism mediated by bacterial phospholipases C (PLCs). A population of *L. monocytogenes* with inefficient LLO activity forms Spacious *Listeria*-containing Phagosomes (SLAPs), which are autophagosome-like compartments that do not mature, allowing slow bacterial growth within enlarged vesicles. SLAPs may represent a stalemate between bacterial LLO action and the host autophagy system, resulting in persistent infection.

Development of a mariner-based transposon and identification of *Listeria monocytogenes* determinants, including the peptidyl-prolyl isomerase PrsA2, that contribute to its hemolytic phenotype.

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Listeriolysin O (LLO) is a pore-forming toxin that mediates phagosomal escape and cell-to-cell spread of the intracellular pathogen *Listeria monocytogenes*. In order to identify factors that control the production, activity, or secretion of this essential virulence factor, we constructed a Himar1 mariner transposon delivery system and screened 50,000 mutants for a hypohemolytic phenotype on blood agar plates. Approximately 200 hypohemolytic mutants were identified, and the 51 most prominent mutants were screened ex vivo for intracellular growth defects. Eight mutants with a phenotype were identified, and they contained insertions in the following genes: lmo0964 (similar to yjbH), lmo1268 (clpX), lmo1401 (similar to ymdB), lmo1575 (similar to ytqI), lmo1695 (mprF), lmo1821 (similar to prpC), lmo2219 (prsA2), and lmo2460 (similar to cggR). Some of these genes are involved in previously unexplored areas of research with *L. monocytogenes*: the genes yjbH and clpX regulate the disulfide stress response in *Bacillus subtilis*, and the prpC phosphatase has been implicated in virulence in other gram-positive

pathogens. Here we demonstrate that prsA2, an extracytoplasmic peptidyl-prolyl cis/trans isomerase, is critical for virulence and contributes to the folding of LLO and to the activity of another virulence factor, the broad-range phospholipase C (PC-PLC). Furthermore, although it has been shown that prsA2 expression is linked to PrfA, the master virulence transcription factor in *L. monocytogenes* pathogenesis, we demonstrate that prsA2 is not directly controlled by PrfA. Finally, we show that PrsA2 is involved in flagellum-based motility, indicating that this factor likely serves a broad physiological role.

Listeria InlB Expedites Vacuole Escape and Intracellular Proliferation by Promoting Rab7 Recruitment via Vps34.

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Rapid phagosomal escape mediated by listeriolysin O (LLO) is a prerequisite for *Listeria monocytogenes* intracellular replication and pathogenesis. Escape takes place within minutes after internalization from vacuoles that are negative to the early endosomal Rab5 GTPase and positive to the late endosomal Rab7. Using mutant analysis, we found that the listerial invasin InlB was required for optimal intracellular proliferation of *L. monocytogenes*. Starting from this observation, we determined in HeLa cells that InlB promotes early phagosomal escape and efficient Rab7 acquisition by the *Listeria*-containing vacuole (LCV). Recruitment of the class III phosphoinositide 3-kinase (PI3K) Vps34 to the LCV and accumulation of its lipid product, phosphatidylinositol 3-phosphate (PI3P), two key endosomal maturation mediators, were also dependent on InlB. Small interfering RNA (siRNA) knockdown experiments showed that Vps34 was required for Rab7 recruitment and early (LLO-mediated) escape and supported InlB-dependent intracellular proliferation. Together, our data indicate that InlB accelerates LCV conversion into an escape-favorable Rab7 late phagosome via subversion of class III PI3K/Vps34 signaling. Our findings uncover a new function for the InlB invasin in *Listeria* pathogenesis as an intracellular proliferation-promoting virulence factor. **IMPORTANCE** Avoidance of lysosomal killing by manipulation of the endosomal compartment is a virulence mechanism assumed to be largely restricted to intravacuolar intracellular pathogens. Our findings are important because they show that cytosolic pathogens like *L. monocytogenes*, which rapidly escape the phagosome after internalization, can also extensively subvert endocytic trafficking as part of their survival strategy. They also clarify that, instead of delaying phagosome maturation (to allow time for LLO-dependent disruption, as currently thought), via InlB *L. monocytogenes* appears to facilitate the rapid conversion of the phagocytic vacuole into an escape-conducive late phagosome. Our data highlight the multifunctionality of bacterial virulence factors. At the cell surface, the InlB invasin induces receptor-mediated phagocytosis via class I PI3K activation, whereas after internalization it exploits class III PI3K (Vps34) to promote intracellular survival. Systematically elucidating the mechanisms by which *Listeria* interferes with PI3K signaling all along the endocytic pathway may lead to novel anti-infective therapies.